

WATKINSVILLE 5 SSE, GA, USA

WMO: 740060

Lat: 33.7837N

Lon: 83.3896W

Elev: 741

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	26.7	7.6	8.3	28.3	11.1	10.0	32.0	14.1	46.8	12.7	46.3	4.3	N/A	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	94.3	74.2	92.2	74.2	90.0	74.0	77.9	87.3	77.1	86.4	76.3	85.5	5.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	136.8	81.2	74.5	132.6	80.5	73.6	128.6	79.9	41.8	87.3	40.9	86.4	40.1	85.9	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.3	8.8	16.4	97.7	4.2	3.4	13.4	100.2	10.9	102.1	8.6	104.1	5.5	106.5		
			14.3	79.4	4.6	1.0	11.0	80.1	8.3	80.7	5.8	81.3	2.5	82.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	43.9	47.5	54.1	61.8	69.9	77.1	79.1	78.5	73.5	63.2	52.6	47.2
	DBStd	14.35	9.69	8.93	8.56	6.92	6.02	3.97	3.30	3.69	4.92	7.48	7.64	8.54
	HDD50	670	242	142	58	6	0	0	0	0	0	4	59	159
	HDD65	2715	654	491	348	138	30	0	0	0	4	122	376	552
	CDD50	5215	53	72	186	361	617	813	901	882	706	413	137	74
	CDD65	1785	0	2	11	43	181	363	436	417	260	66	4	2
	CDH74	15661	0	5	97	438	1613	3358	4015	3607	2011	492	23	2
	CDH80	6308	0	0	8	67	526	1475	1801	1576	749	106	0	0
Wind	WSAvg	4.0	4.6	4.7	4.9	4.3	3.8	3.5	3.3	3.1	3.3	3.7	3.9	4.3
Precipitation	PrecAvg	49.2	4.4	4.5	4.9	3.5	3.3	4.4	4.4	4.4	4.1	3.1	3.7	4.4
	PrecMax	65.1	7.0	8.5	9.3	9.3	8.9	11.7	10.1	9.5	12.8	9.2	9.4	12.2
	PrecMin	32.3	2.4	1.7	1.2	1.3	1.0	1.3	1.4	1.0	0.0	0.0	0.7	1.9
	PrecStd	9.0	1.3	1.6	2.0	1.9	1.6	2.4	2.5	2.3	2.8	2.2	2.1	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.8	74.3	81.1	85.2	92.5	96.4	96.1	97.7	92.9	87.9	77.4	72.7
		MCWB	60.5	63.8	62.8	66.8	70.2	73.6	75.0	73.4	72.6	71.2	64.2	63.8
	2%	DB	66.5	69.2	76.8	81.9	88.6	93.3	93.6	93.8	90.2	82.5	73.0	68.9
		MCWB	58.0	59.7	61.9	66.1	69.5	74.1	75.3	73.8	72.4	67.8	60.3	63.3
	5%	DB	63.2	65.9	72.9	79.2	85.7	90.8	91.6	90.8	87.6	79.6	69.5	65.3
		MCWB	57.0	58.3	59.9	64.3	69.4	74.0	75.5	74.5	72.1	66.2	59.8	60.7
	10%	DB	59.6	62.4	69.1	76.3	82.7	88.2	89.2	88.2	84.7	76.1	66.2	61.7
		MCWB	54.2	56.0	57.4	62.8	68.8	73.6	75.1	74.4	71.6	64.9	58.3	57.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	66.0	66.1	70.9	74.3	78.2	78.8	78.8	76.9	75.0	68.5	68.2
		MCDB	67.9	71.0	75.3	79.0	83.3	88.3	88.8	87.9	86.1	82.1	71.7	70.5
	2%	WB	61.5	63.9	64.4	68.8	72.8	77.0	77.9	77.8	75.4	72.8	65.5	65.1
		MCDB	65.1	68.5	73.0	77.3	82.1	87.6	88.1	86.1	85.2	78.4	69.5	67.3
	5%	WB	58.7	61.1	62.5	67.3	71.6	75.9	77.2	77.0	74.4	70.5	62.7	61.6
		MCDB	61.7	65.2	70.2	75.2	80.6	86.8	87.2	85.0	83.2	76.1	66.0	63.9
	10%	WB	54.8	58.2	60.5	65.5	70.5	74.9	76.3	76.0	73.2	67.9	59.6	57.6
		MCDB	58.0	61.6	66.8	73.5	79.1	85.2	86.0	83.9	80.9	73.8	64.3	60.9
Mean Daily Temperature Range	5% DB	MDBR	18.8	19.2	20.9	22.3	21.7	20.1	19.2	18.6	19.2	20.8	20.9	17.8
		MCDBR	21.0	21.5	24.2	25.3	25.3	23.8	22.3	22.6	23.3	24.0	24.4	20.0
		MCWBR	14.6	12.8	11.9	11.3	9.5	7.8	6.7	7.5	8.3	10.7	14.0	13.1
	5% WB	MCDBR	17.8	17.0	20.1	20.3	19.9	20.2	19.4	18.0	19.1	18.8	17.3	15.4
		MCWBR	15.1	12.4	11.4	10.9	8.7	7.9	7.3	6.8	7.6	10.0	12.9	12.9
Clear-Sky Solar Irradiance	taub	0.314	0.319	0.355	0.389	0.450	0.480	0.520	0.518	0.423	0.383	0.342	0.311	
	taud	2.521	2.499	2.425	2.317	2.185	2.150	2.058	2.049	2.299	2.390	2.456	2.555	
	Ebn at Noon	286	296	292	285	267	258	247	245	266	269	272	281	
	Edh at Noon	26	30	35	41	47	49	53	53	39	33	28	24	
All-Sky Solar Radiation	RadAvg	870	1057	1400	1746	1931	1990	1892	1735	1508	1274	964	741	
	RadStd	48	127	101	116	153	152	128	88	145	150	93	76	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+203

Nomenclature: See separate page